

LESSON 2.3

REWRITING EXPRESSIONS USING THE DISTRIBUTIVE PROPERTY



LESSON INTRODUCTION

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

LEARNING TARGETS

- I can use the distributive property to generate equivalent numeric expressions.

BENCHMARK COHERENCE

WORKING TOWARD

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

ESSENTIAL QUESTIONS

- How can the distributive property be used to generate equivalent expressions?
- What are the similarities and differences between writing equivalent expressions using the area model and the distributive property?

LESSON NARRATIVE

Students will extend prior knowledge of the distributive property to area models for rewriting numeric expressions. The lesson includes rewriting multiplication expressions and addition expressions using the distributive property and factors to build stronger foundations of equivalent numeric expressions before delving deeper into rewriting sums of composite numbers with a common factor in the next lesson.

COMPONENT SUMMARY

2.3.1 WARM-UP (~5 MIN.)

Practice with using factors for rewriting expressions

2.3.2. EXPLORATION (5-10 MIN.)

Students connect the area model to distributive property

2.3.3 GUIDED INSTRUCTION (10-15 MIN.)

Guiding prompts for applying the area model for representing distributive property

2.3.4 GUIDED PRACTICE (5-15 MIN.)

Practice rewriting expressions using the distributive property and area model

2.3.5 YOUR TURN! (10 MIN.)

Practice writing equivalent expressions and evaluating an error in the rewriting process

2.3.6 WRAP-UP (~5 MIN.)

Students complete statements for a growth mindset self-assessment on this lesson

RESOURCES

GLOSSARY

Key Terms:

- Distributive property of multiplication over addition

Additional Terms:

- N/A

MATERIALS

- (Optional) Visual anchor chart for distributive property as area model

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with fluency in identifying factors and dividing by factors.
- Students with gaps in conceptual understanding of area may have difficulty using the area model to show the distributive property.
- Students may struggle to differentiate between expressions and equations.

THINGS TO CONSIDER

- Consider focusing on discussion of the distributive property to rewrite expressions. The next lesson will provide additional practice rewriting addition expressions using common factors, so discussion in this lesson can prioritize conceptual understanding.
- Throughout the lesson, math vocabulary is used to identify parts of an expression and answers. Consider reviewing the vocabulary words sum, product, addend, and factor before or during instruction.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

2.3.2 Exploration positions students to make connections among the various representations of equivalent expressions including the area model, distributive property of multiplication over addition, and product of three factors. Consider ways to highlight these connections throughout the lesson, like highlighting in interactive notebooks or tallying terms.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

2.3.5 Your Turn! and 2.3.6 Wrap-Up cultivate a growth mindset by analyzing mistakes and reflecting on gaps or questions in learning. Teachers should frame perseverance by modifying methods as needed when solving a challenging task. Teachers can also model how to maintain a positive mindset when working to solve tasks through Guided Instruction and explicitly using growth-mindset language.

DIFFERENTIATE INSTRUCTION

Notice additional entry points for distributive property in 2.3.2 Exploration. Students engage with a visual model of the distributive property – consider manipulatives or sketching tools for students to make sense of the distributive property through this lens.

Consider the different entry points and ways to engage with approaches for rewriting multiplication and addition expressions. Consider creating a T-chart to organize and provide examples of different strategies, using diagrams and numeric expressions as an entry points for visual learners.

2.3.6 Wrap-Up is an open-ended growth mindset self-assessment. For students who may struggle to write using sentence stems, consider giving an opportunity to draw or create a diagram.

2.3.1 WARM-UP: HOW MANY WAYS?

Component Overview: Students first individually generate various ways to express 120 as a product of three factors, then share with a partner to think of different or new equivalent expressions. Give students an opportunity to discuss and share ideas with their partner.

TEACHER MOVES

Think-Pair-Share

Consider fostering partner and whole-group discussions on the various arrangements and factors used, focusing on equivalent expressions equal to 120. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as “The factors I chose were...because...” or “These expressions are equivalent because...”



2.3.1 Warm-Up: How Many Ways?

- Without using the number 1 as a factor, determine different ways the number 120 can be written as a product of three factors. Find at least three different expressions.
Answers will vary. $2 \cdot 3 \cdot 20$; $8 \cdot 3 \cdot 5$; $4 \cdot 6 \cdot 5$
- Work with your partner to share the different ways 120 can be expressed as a product of three factors. Add any new expressions shared to your list above and discuss similarities and differences between the expressions you and your partner determined.



2.3.2 Exploration: The Distributive Property

A local grocery store manager is trying to determine how many soda cans she will need to make a new display. The specifications require a total of 7 rows with 23 cans in each row.

To determine the number of cans needed, she multiplies the number of rows by the number of cans in each row.

1. What is the value of 7×23 ? Show your work.
 161

2. Compare your work with two other students.
 Answers will vary.

Name	Did you get the same answer?	Did you use a similar method? If not, summarize their method below.

3. a. Find the products.

7×20
140

7×3
21

- b. What is the sum of the two products?
 161

4. How does the sum of those two products compare to the product of 7×23 ?
- The sum of $(7 \times 20) + (7 \times 3)$

product of 7×23 .

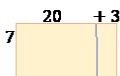
○ is less than
 ○ is equal to
 ○ is greater than

the

Area models can be helpful when finding products of numbers. The figure to the right can be used to model the product of 7×23 .



The number 23 can also be decomposed into $(20 + 3)$ to find the same product. In this area model, both parts of the area model can be added to find the total area.



5. How can decomposing the number 23 into $20 + 3$ make multiplying easier?
 Answers will vary. By decomposing numbers, it allows you to use mental math to solve the problem.

2.3.2 EXPLORATION: THE DISTRIBUTIVE PROPERTY

Component Overview: This Exploration guides students through generating equivalent expressions using the distributive property by starting with multiplicative relationships and building toward rewriting sums with the distributive property. The area model is used to help students find products of numbers using the distributive property.

DIFFERENTIATED INSTRUCTION

As an entry point for visual learners, consider having students draw what they think this soda can display would look like. Use circles or ovals to represent the cans. Revisit this drawing after students complete questions 3-5 to help make the connection between this array and the area model.

DIFFERENTIATED INSTRUCTION

Consider allowing students to walk around the room to find partners as they complete the table in question 2.

TEACHER MOVES

Consider highlighting student responses that use different strategies to find the product of 7×23 . This discourse can position students to better make connections for writing various equivalent numeric expressions in different ways in Lesson 4.

ENRICHMENT

How does the area model show the distributive property? What value is being distributed?

TEACHER MOVES

Compare and Connect

Have students make connections to their responses to the first part of this exploration. Ask students to notice if their responses to question 3 are equivalent to the expression represented in the area model. Students who may not have made the connection to 7 as a common factor being distributed to $20 + 3$ might now better connect how to rewrite equivalent expressions using the distributive property.

2.3.3 GUIDED INSTRUCTION: THE DISTRIBUTIVE PROPERTY

Component Overview: Students are introduced to the definition of the distributive property and extend their understanding to addition expressions. Instruction also focuses on flexibility to rewrite expressions using different addends within the parentheses. Students also practice using expressions written with the distributive property to find products using mental math.

TEACHER MOVES

Consider building vocabulary and conceptual understanding of the distributive property by using the word “distribute” in a non-math context. Connect “distribute” to “give” with statements like, “If I distribute candy, that will mean EVERY person in this room would get a piece of candy. To distribute means to give.”

QUESTIONING

Have students build upon question 5 from 2.3.2 Exploration by practicing mental math to find the product of an expression using the distributive property. For example, highlight that using the decomposed factors to multiply 7×20 and 7×3 , then adding the products, is easier than multiplying 7×23 mentally. Challenge students to generate another example and explain how mental math and distribution arrive at the same answer.

DIFFERENTIATED INSTRUCTION

As an entry point for visual learners to “see” the distributive property, consider drawing arrows from the factor outside the parentheses to the addend inside the parentheses as you multiply to rewrite equivalent expressions as the sum of two products. Emphasize that the value is multiplied by both addends.

ENRICHMENT

When rewriting sums, how can you determine which factor to distribute? Does the factor need to be the greatest common factor?



2.3.3 Guided Instruction: The Distributive Property



The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$7 \times 23 = 7(20 + 3) \\ 7(20 + 3) = (7 \cdot 20) + (7 \cdot 3)$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

Answers will vary.

$$7 \times 23 = 7(\underline{10} + \underline{13}) \\ 7(23) = 7(\underline{11} + \underline{12})$$

2. Which of the expressions is the easiest to evaluate using mental math?

Answers will vary. $7(10 + 13)$; $7(20 + 3)$.

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.

3. Elase rewrote the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

- a. $27 + 18$ and $3(9 + 6)$ ☐ are ☐ are not equivalent

expressions because when evaluated they have

- ☐ the same value.
☐ different values.

- b. What common factor of 27 and 18 did Elase use to rewrite the expression?

3

- c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.

$$\begin{array}{c} \underline{9} \quad \begin{array}{|c|c|} \hline \underline{3} & \underline{2} \\ \hline 27 & 18 \\ \hline \end{array} \\ 27 + 18 = \underline{9}(\underline{3} + \underline{2}) \end{array}$$



2.3.4 Guided Practice: Rewriting Expressions

1. Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5(90 + 8)$	5 90 + 8 45040	490
6×22	Answers will vary. $6(12 + 10)$	6 12 + 10 7260	132

2. Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum						
600 + 140	Answers will vary. $20(30 + 7)$	20 <table><tr><td>30</td><td>+</td><td>7</td></tr><tr><td>600</td><td></td><td>140</td></tr></table>	30	+	7	600		140	740
30	+	7							
600		140							
56 + 32	Answers will vary. $8(7 + 4)$	8 <table><tr><td>7</td><td>+</td><td>4</td></tr><tr><td>56</td><td></td><td>32</td></tr></table>	7	+	4	56		32	88
7	+	4							
56		32							

2.3.4 GUIDED PRACTICE: REWRITING EXPRESSIONS

Component Overview: Students synthesize learning of rewriting multiplication expressions in three ways: using the distributive property, modeling with area, and finding the product.

TEACHER MOVES

The second row of each table has multiple correct answers using various equivalent expressions. Encourage discourse on how various factors, not just the GCF, can be used to rewrite the expression using the distributive property. Use math vocabulary like “equal to” and “equivalent expressions” to highlight that the expressions are still equivalent.

ENRICHMENT

When finding the product for these expressions, how can decomposing factors make it easier to multiply using mental math?

QUESTIONING

Consider an error analysis for the distributive property expressions in the question 2 table. For example, look at student work to identify a “My Favorite No.” Use this opportunity to emphasize procedural knowledge and a growth mindset by learning from mistakes.

2.3.5 YOUR TURN!

Component Overview: Students summarize and synthesize learning by using the distributive property to rewrite both multiplication and addition numeric expressions. Students then analyze a mistake in writing equivalent expressions, correct the mistake, and suggest ways to learn from the mistake.

ENRICHMENT

What does it mean for two expressions to be equivalent? How can you determine whether the expressions in each row are equivalent?

TEACHER MOVES

Consider using question 2 to emphasize how making mistakes is an important part of learning. Use questioning prompts to encourage discussion on how to foster a growth mindset within the context, including “Why is it important that Bailey reflects on his mistake before moving on?”



2.3.5 Your Turn!

- For each expression, generate an equivalent numeric expression using the distributive property and then find the value. **Answers will vary for equivalent expressions.**

Expression	Equivalent Expression Using Distributive Property	Value
9×16	$9(10 + 6)$	144
$25 \cdot 8$	$8(20 + 5)$	200
39×4	$4(30 + 9)$	156
$48 + 72$	$8(6 + 9)$	120
$21 + 98$	$7(3 + 14)$	119

- Bailey rewrote the expression $36 + 81$ using the distributive property, but he made an error.

a. Evaluate his work and determine his mistake.

Given Expression	Bailey's Expression	Corrected Expression
$36 + 81$	$9(4 + 81)$	$9(4 + 9)$

- Help Bailey understand and learn from his mistakes. Write a brief note to Bailey explaining what his mistake was and how he can learn from it.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Answers will vary. Always divide both values in the parentheses by the common factor.

**2.3.6 Wrap-Up: Growth Mindset**

1. Complete each statement below based on your learning today.

Answers will vary.

Today I am still unsure about _____
_____.

To fill this gap, I plan to _____
_____.

2.3.6 WRAP-UP: GROWTH MINDSET

Component Overview: This Wrap-Up provides an opportunity for reflection and cultivating a growth mindset in learning mathematics. The self-assessment fosters metacognition as students consider what concepts they are still unsure of and create a plan to complete this learning.

DIFFERENTIATED INSTRUCTION

For auditory learners, consider allowing each student to read their completed sentences to you or record themselves reading their growth mindset self-assessment. For visual learners, consider allowing students to draw or create a picture in the space below each question.